Business Solutions to the Coffee Crisis

TECHNOSERVE

Business Solutions to Rural Poverty

December 2003

EXECUTIVE SUMMARY

Producers of coffee and their employees continue to endure a profound economic and humanitarian crisis. For many of the world's coffee producers, coffee prices remain lower than the cost of production for the third straight season.

The coffee industry has experienced crises in the past; however, this crisis is unique. There has been a systemic shift in the cost structure of the coffee industry as a result of innovation in Brazil and the growth of Vietnam as low-cost producer. While prices may recover in the short term, long-term prices are likely to remain in a range that will leave many producers unprofitable. Many workers on medium and large-sized farms will be particularly vulnerable in this altered competitive landscape.

While there is no single answer to this crisis, TechnoServe believes there are three market solutions which offer the highest potential for sustainable impact:

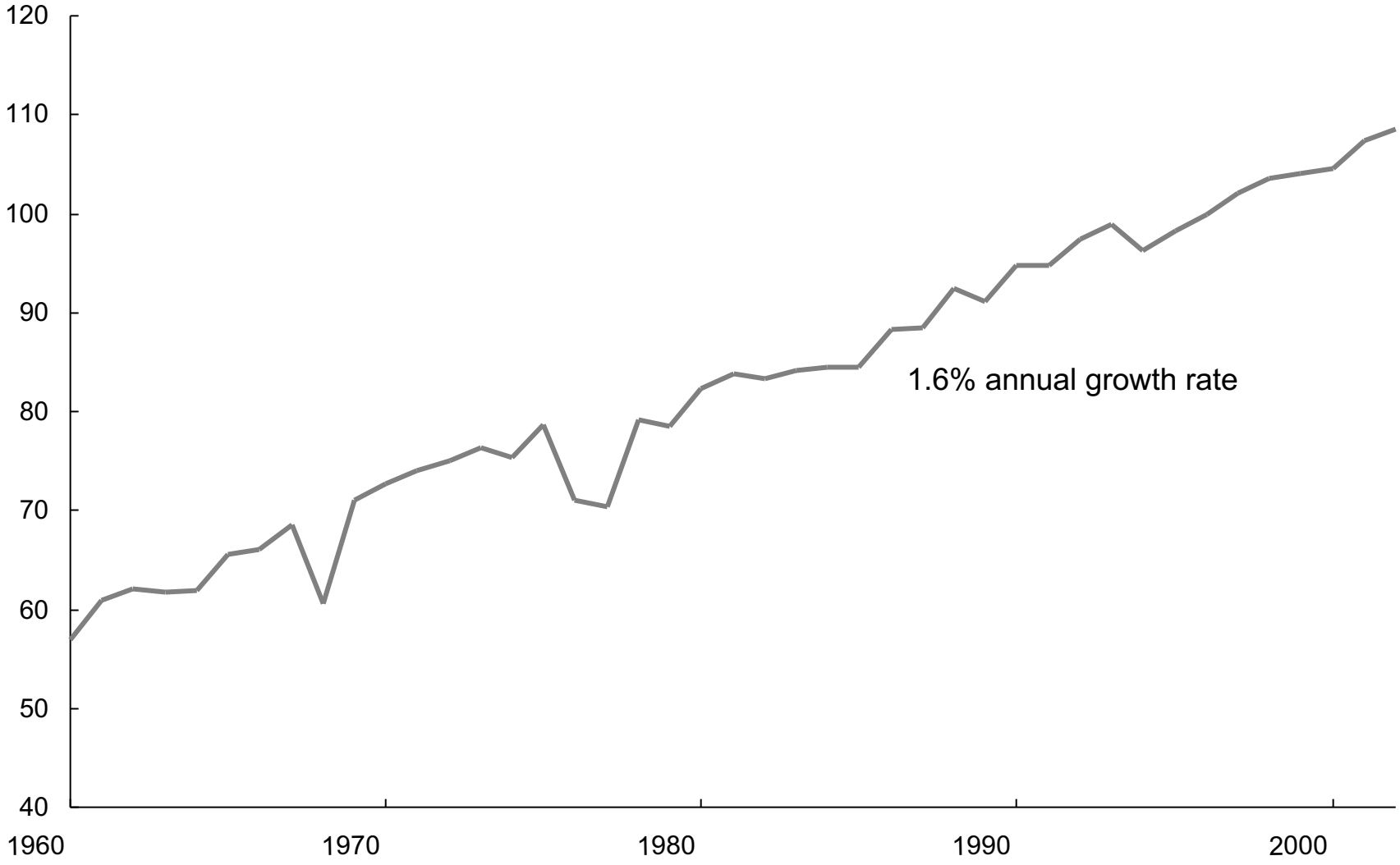
- Increase coffee consumption in producing countries and emerging market countries
- Support specialty coffee producers in their efforts to secure market premiums
- Promote diversification into other products and industries for regions with a high concentration of marginal coffee producers that do not have the potential to differentiate via specialty markets.

CONTENTS

- **Overview of the Current Crisis**
- Likely Developments
- Potential Solutions
- Next Steps

COFFEE CONSUMPTION HAS GROWN STEADILY OVER THE PAST 40 YEARS

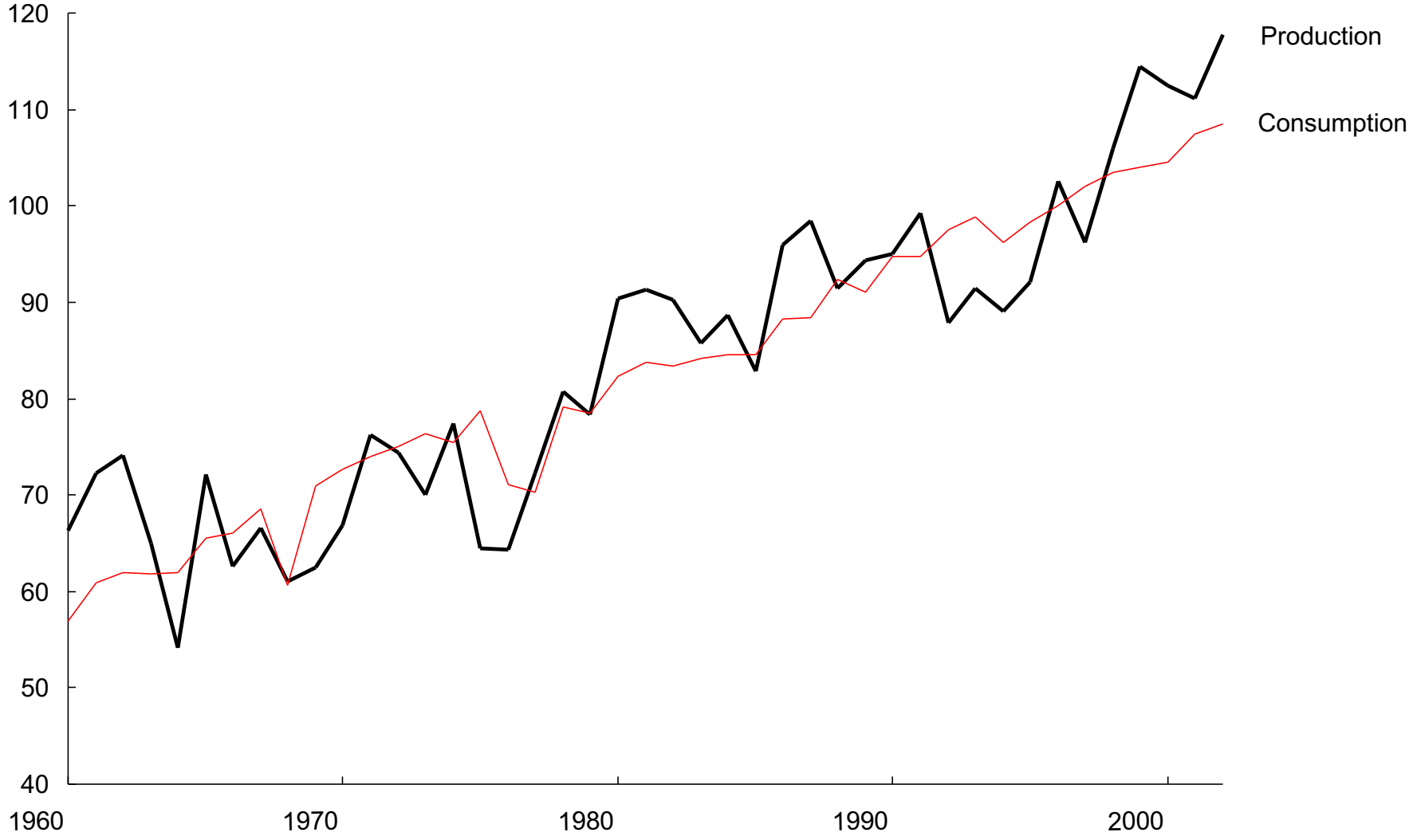
Green coffee consumption; millions of 60 kg bags



Source: ICO; World Bank; *150 Years of Coffee*, Marcellino Martins and E. Johnston

COFFEE PRODUCTION HAS BEEN SUBJECT TO MORE VOLATILITY THAN CONSUMPTION

Green coffee; millions of 60 kg bags



Source: ICO; World Bank; *150 Years of Coffee*, Marcellino Martins and E. Johnston

THIS PRODUCTION VOLATILITY HAS BEEN DRIVEN BY TWO FACTORS

Weather

- Brazil's dominant position in world coffee production makes global supply highly susceptible to frosts and drought in Brazil's coffee regions

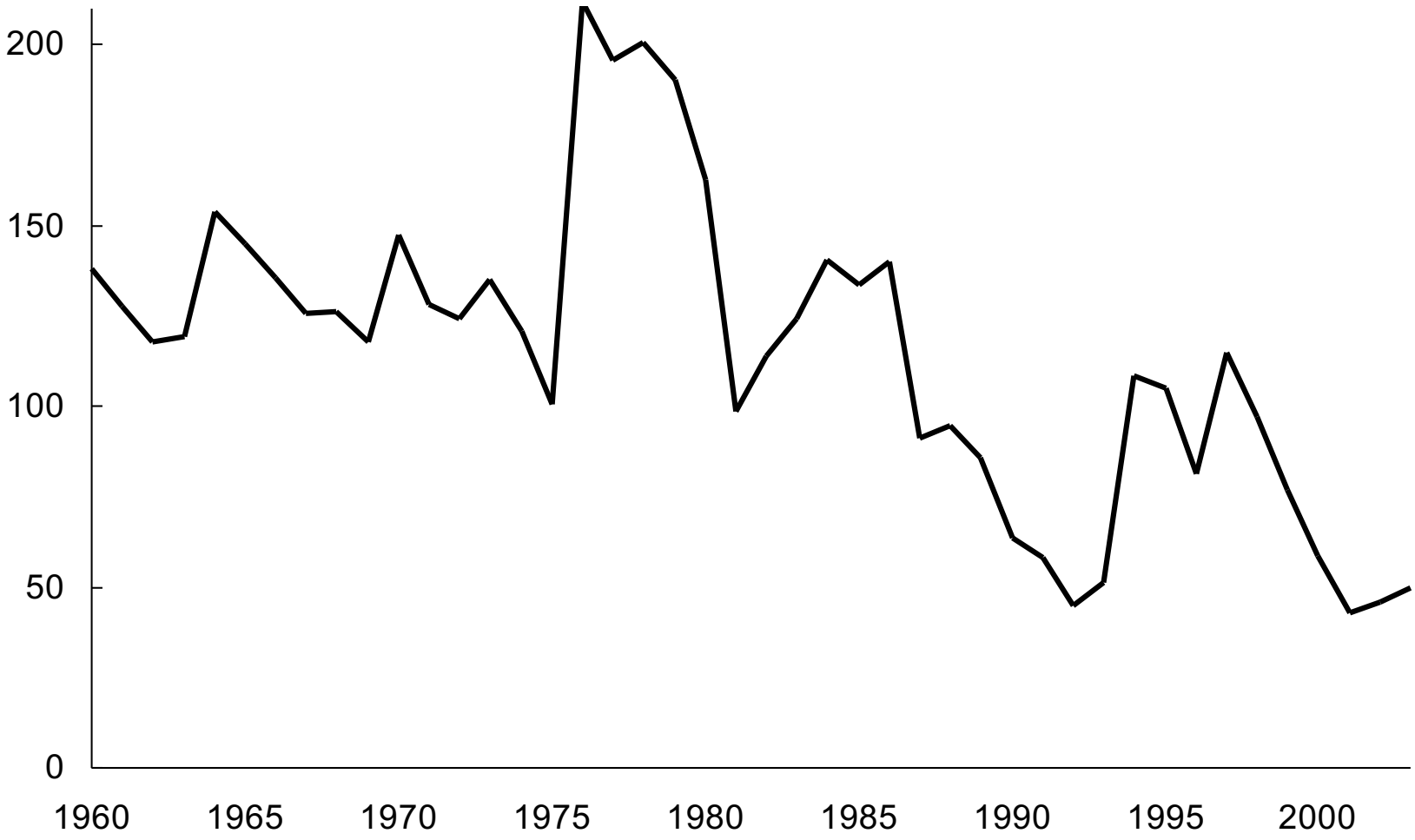
Supply Adjustment Lags

- Supply does not immediately decline when prices drop because coffee must still be picked in order to maintain healthy trees regardless of the market price
- New trees do not start producing fruit for 3 years, which causes a significant lag in production increases related to market events

Source: Interviews; World Bank; team analysis

RECENT PRODUCTION SURPLUSES HAVE CAUSED PRICES TO DROP TO RECORD LOWS

Real price of green coffee
ICO indicator price, U.S. ¢ per lb., constant 2002 dollars*

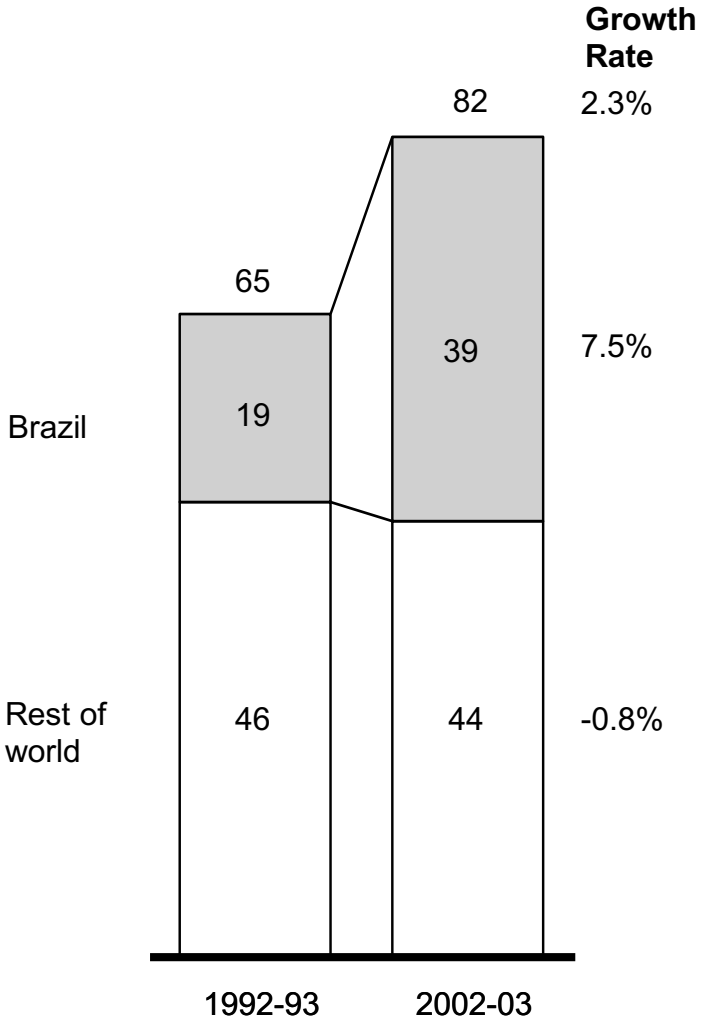


* Real price index created by combining 1900-91 average U.S. import prices from *150 Years of Coffee* and 1992-2002 ICO indicator prices; all prices in 2002 dollars adjusted using World Bank Global Manufactured Unit Value index

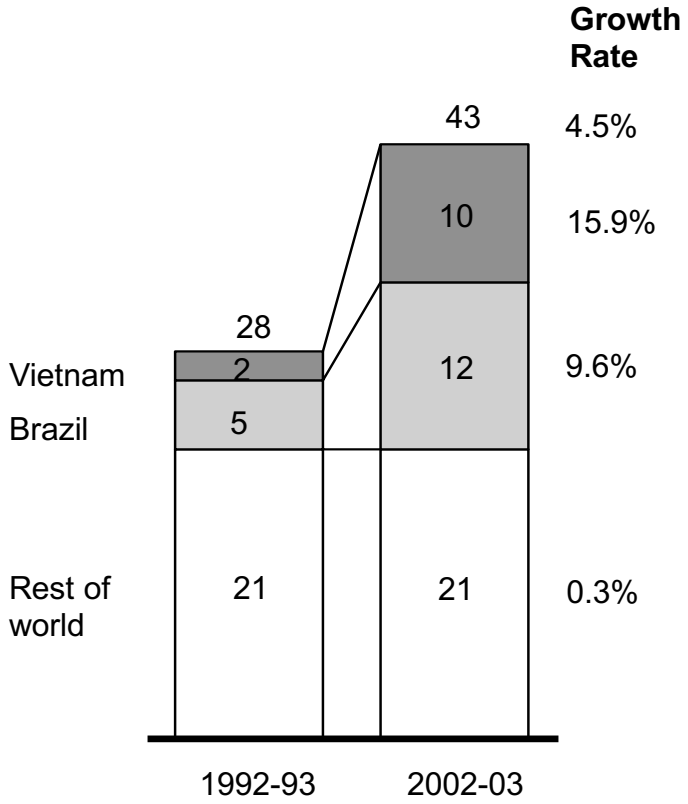
Source: ICO; World Bank; *150 Years of Coffee*, Marcellino Martins and E. Johnston

THE RECENT INCREASE IN PRODUCTION HAS BEEN DRIVEN LARGELY BY BRAZIL AND VIETNAM

***Arabica* production**
(millions of 60 kg bags)

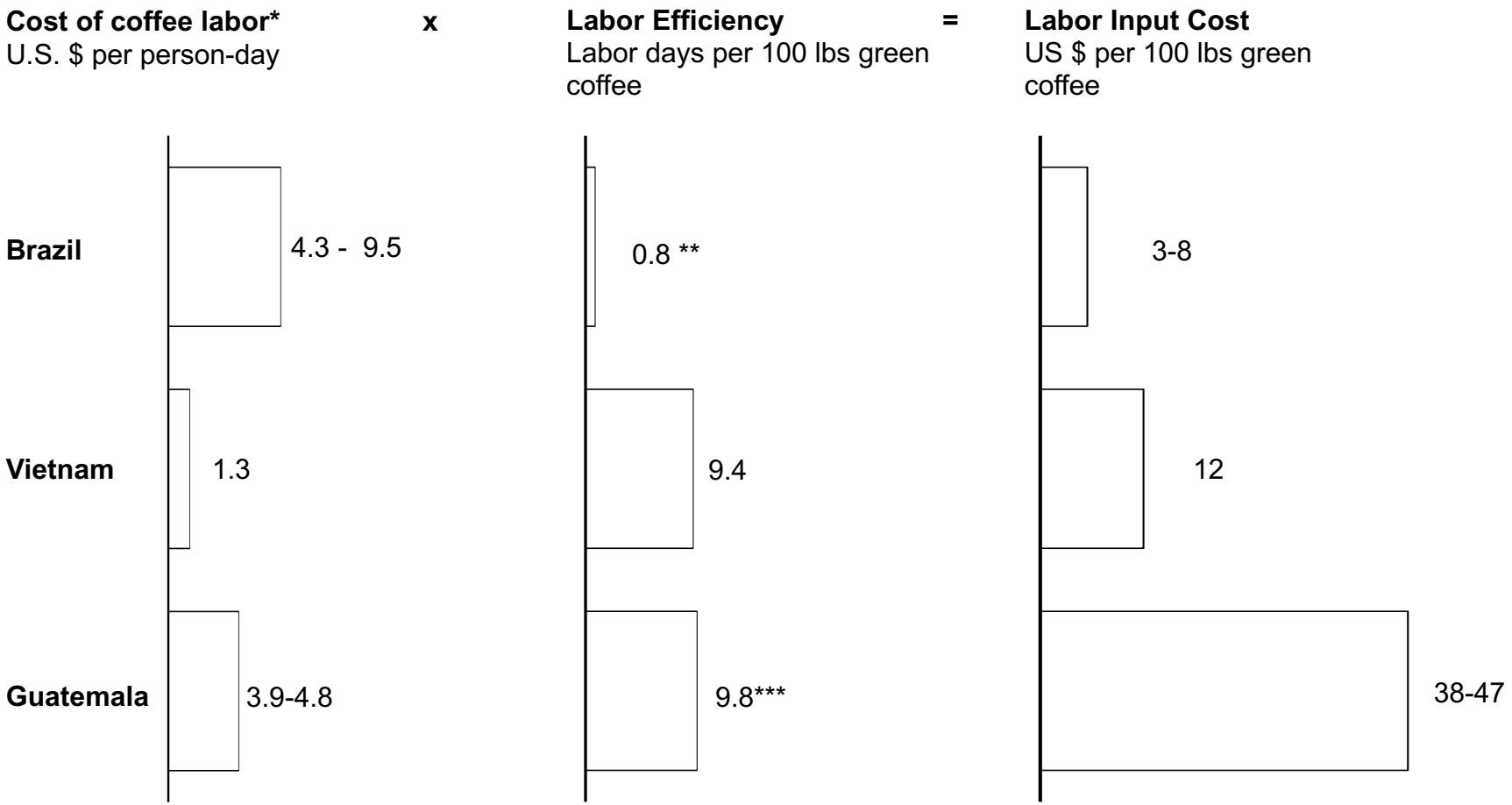


***Robusta* production**
(millions of 60 kg bags)



Source: USDA; interviews

BRAZIL AND VIETNAM HAVE A COST ADVANTAGE DUE TO HIGH EFFICIENCY AND LOW LABOR COSTS



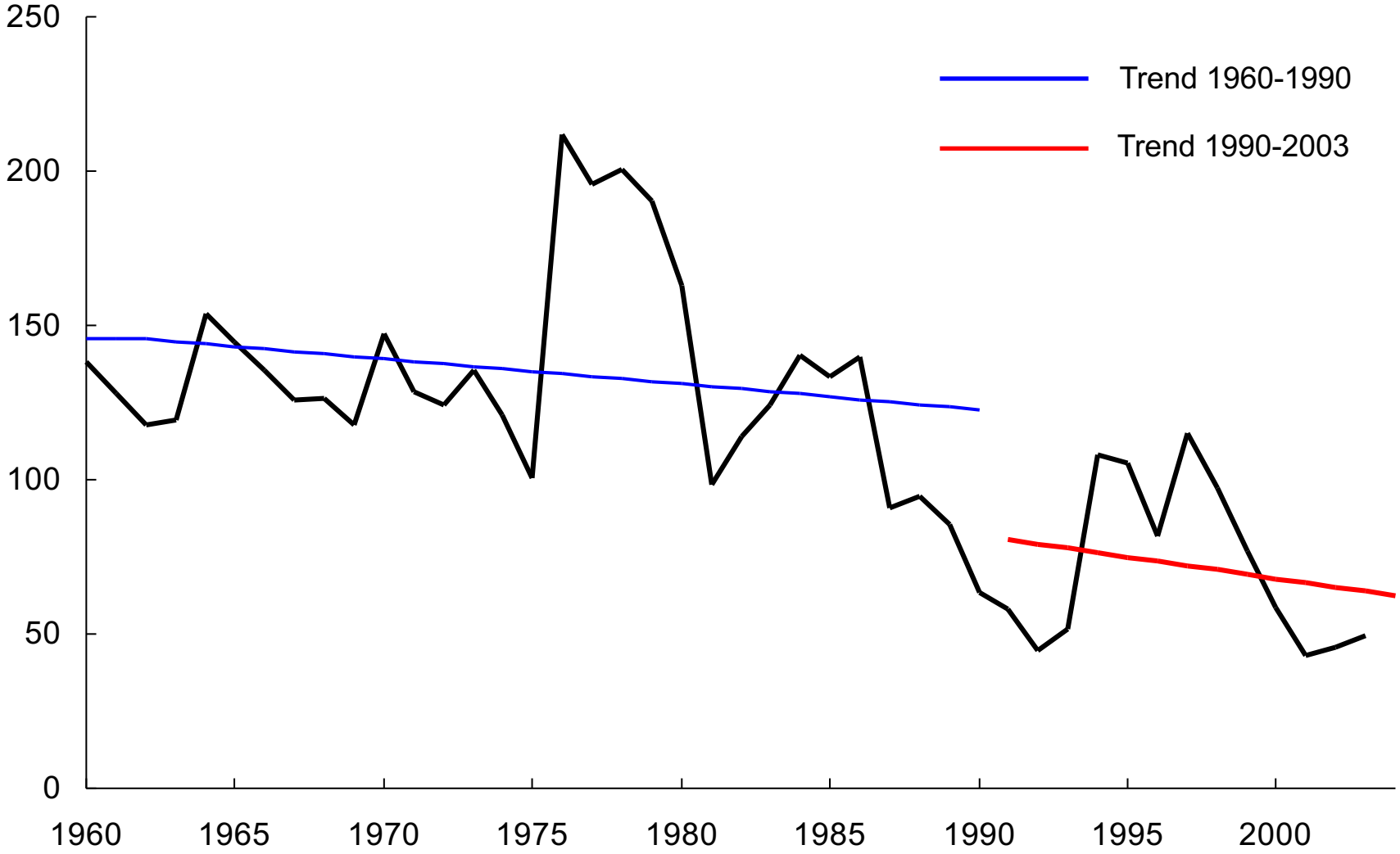
* Range based on direct wages (low-end) and fully loaded cost (high-end)
** Assuming mechanical harvesting
*** Uses Guatemala wage of $4.75 per day and includes wet-milling in labor costs

Source: Interviews, cost models, team analysis

THE PRICE DROP REFLECTS A SYSTEMIC CHANGE IN THE COFFEE INDUSTRY

Price trend green coffee 1960-90 and 1990-2003

ICO indicator price, U.S. ¢ per lb., constant 2002 dollars*



Source: ICO; World Bank; Team Analysis

THREE FACTORS HAVE DRIVEN THIS SYSTEMIC CHANGE SINCE THE END OF THE INTERNATIONAL COFFEE AGREEMENT

Innovation in Brazil

- Migration of production to less frost-prone areas and increased irrigation
- Improved mechanical harvesting
- Government programs providing price forecasting and risk management
- Industry organization across the value chain to increase productivity, coordination and market responsiveness

Entry of Vietnam

- Production has increased from 2 million to 10 million bags in 10 years
- Low labor costs
- Proactive government policies providing yield improvement and market information
- Excess production capacity equivalent to 8% of 2002 global *Robusta* demand

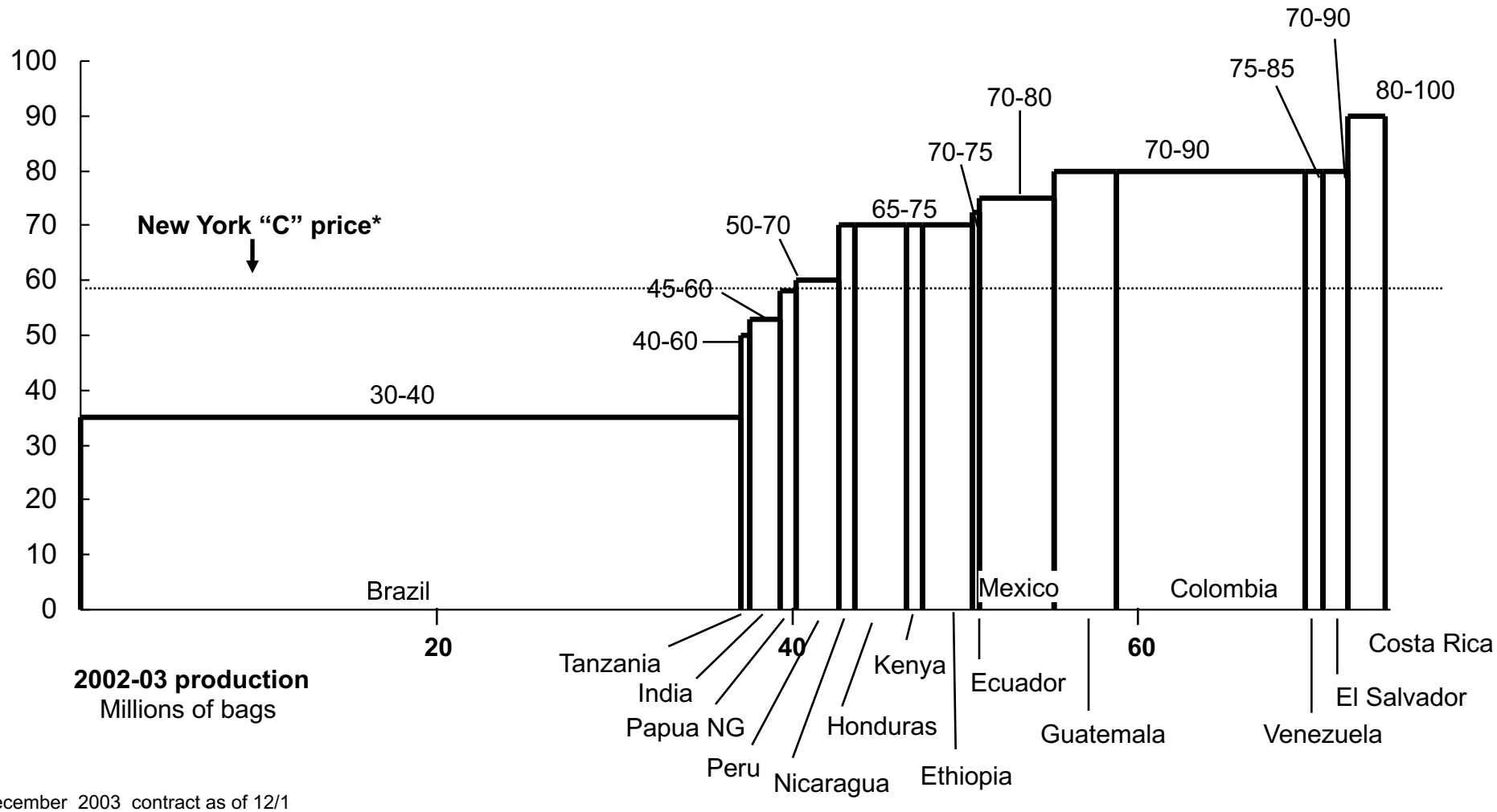
Increasing share of *Robusta*

- New ways to reduce the acidity of *Robusta* and increase its use in blends
- Use of *Robusta* in popular flavored coffees which mask the harsh taste
- Growth of coffee consumption in more price-sensitive countries

Source: Interviews; World Bank; team analysis

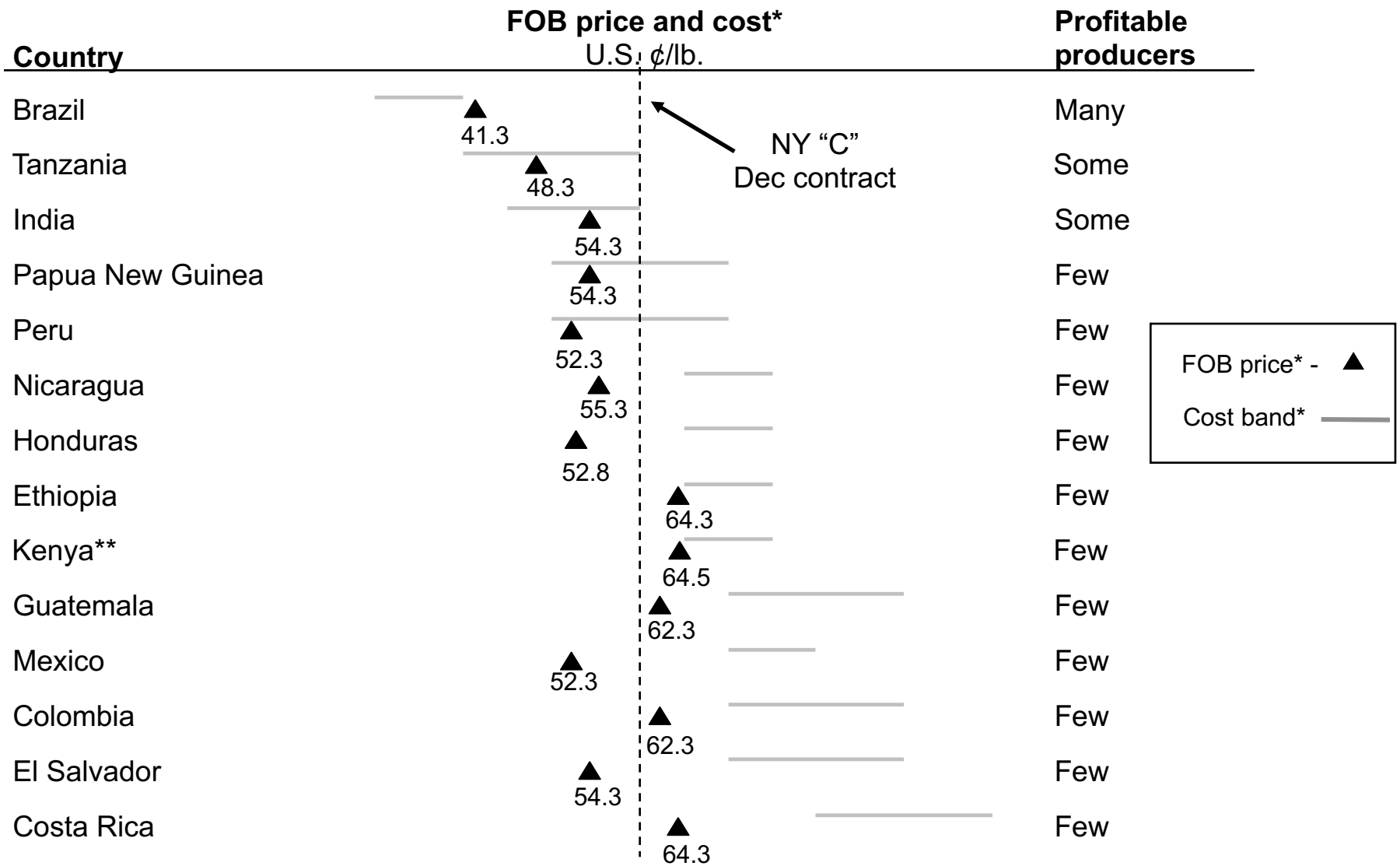
AT CURRENT PRICE LEVELS MANY PRODUCERS IN HIGHER COST COUNTRIES ARE NOT PROFITABLE

***Arabica* FOB supply curve**
Average cost (U.S. ¢/lb.), 1996-2001



* December 2003 contract as of 12/1
Source: ICO; USDA; TechnoServe; Volcafe; Judith Ganes Consulting; Coffee Business International; "Dealing with the Coffee Crisis in Central America," World Bank (2003); team analysis

AND REGIONAL PRICE PREMIUMS DO NOT MAKE UP FOR HIGHER COSTS



* Price based on 12/01/03 New York "C" (59.6 c/lb) +/- current differential (differentials based on Complete Coffee Coverage and other industry estimates); costs are estimates based on data from compilation of sources; Tanzanian costs and price are weighted averages of small and larger growers based on country deep dives

** Cost estimated based on Ethiopia's cost

Source: ICO; TechnoServe; Volcafe; Judith Ganes Consulting; Coffee Business International; "Dealing with the Coffee Crisis in Central America," World Bank (2003); June 30, 2003 *Complete Coffee Coverage*; team analysis

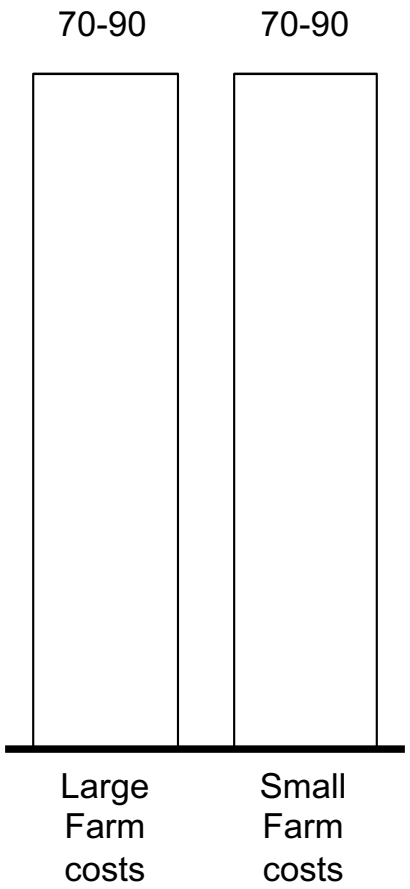
LARGE FARMS AND THEIR WORKERS HAVE BEEN PARTICULARLY HARD HIT

(cents per lb)

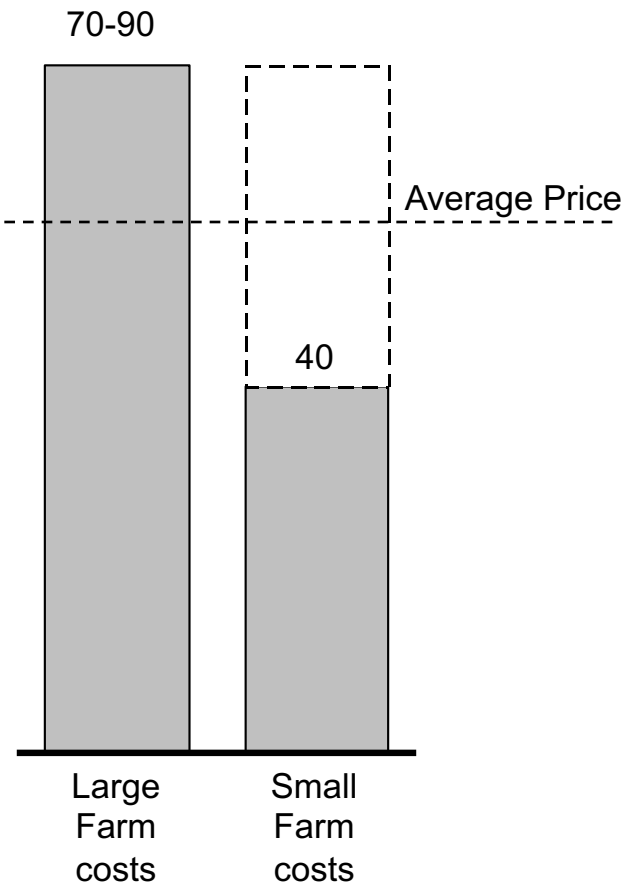
GUATEMALA EXAMPLE

Cash Expenses

There are few economies of scale * …



and large farms do not have access to non-cash family labor…

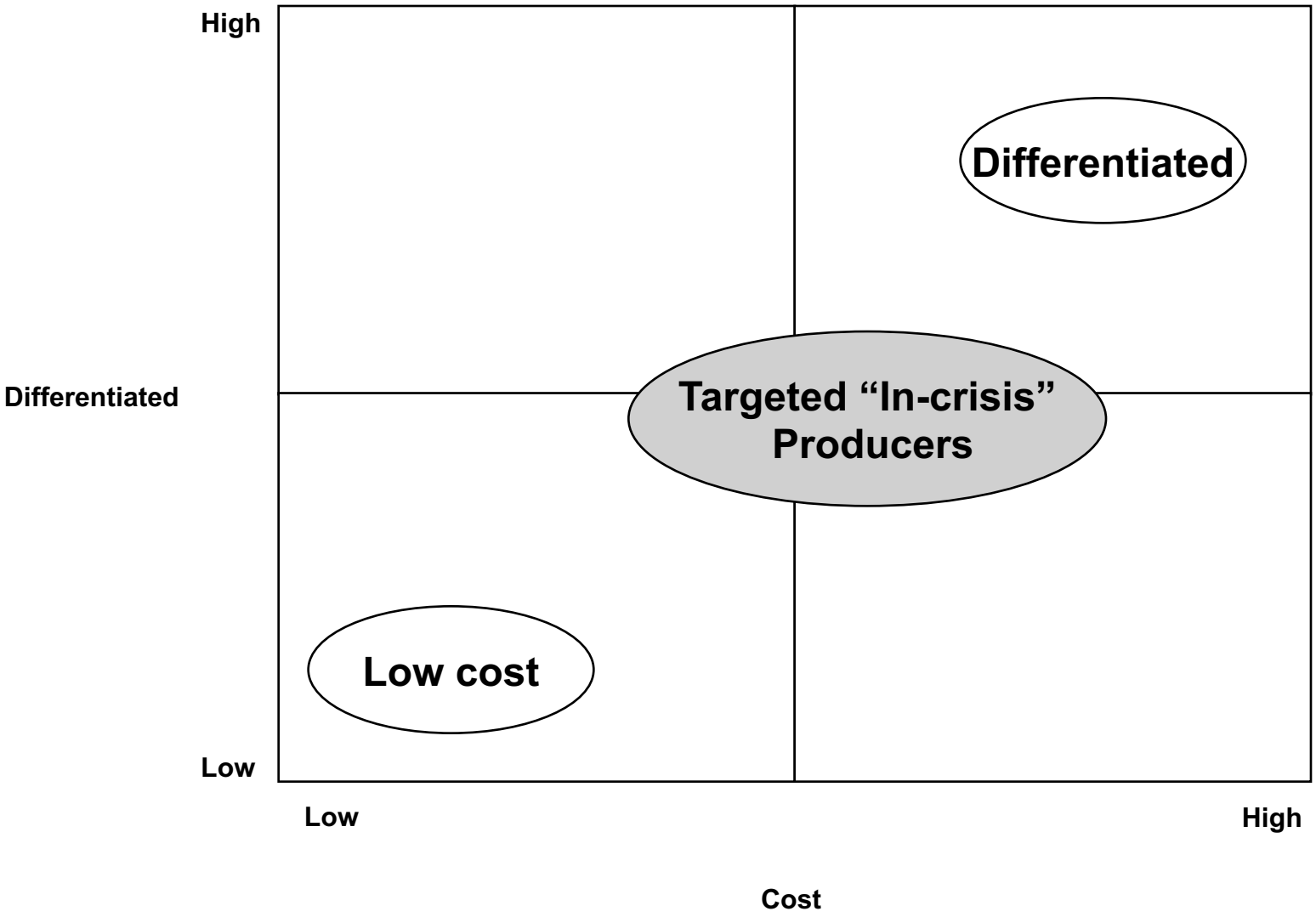


- Larger farms pay cash wages to workers rather than having access to non-cash family labor
- Higher cash expenses make it more difficult for larger farms to withstand price reductions
- 250,000 coffee workers in Guatemala have been laid off -- 30% of the coffee industry work force

* In hand harvesting, the use of mechanized harvesting (Brazil) does offer scale economies

Source: ICO; Unitrade; ANACAFE; team analysis

MARGINAL PRODUCERS THAT CANNOT DIFFERENTIATE OR COMPETE ON COST ARE UNLIKELY TO RETURN TO PROFITABILITY



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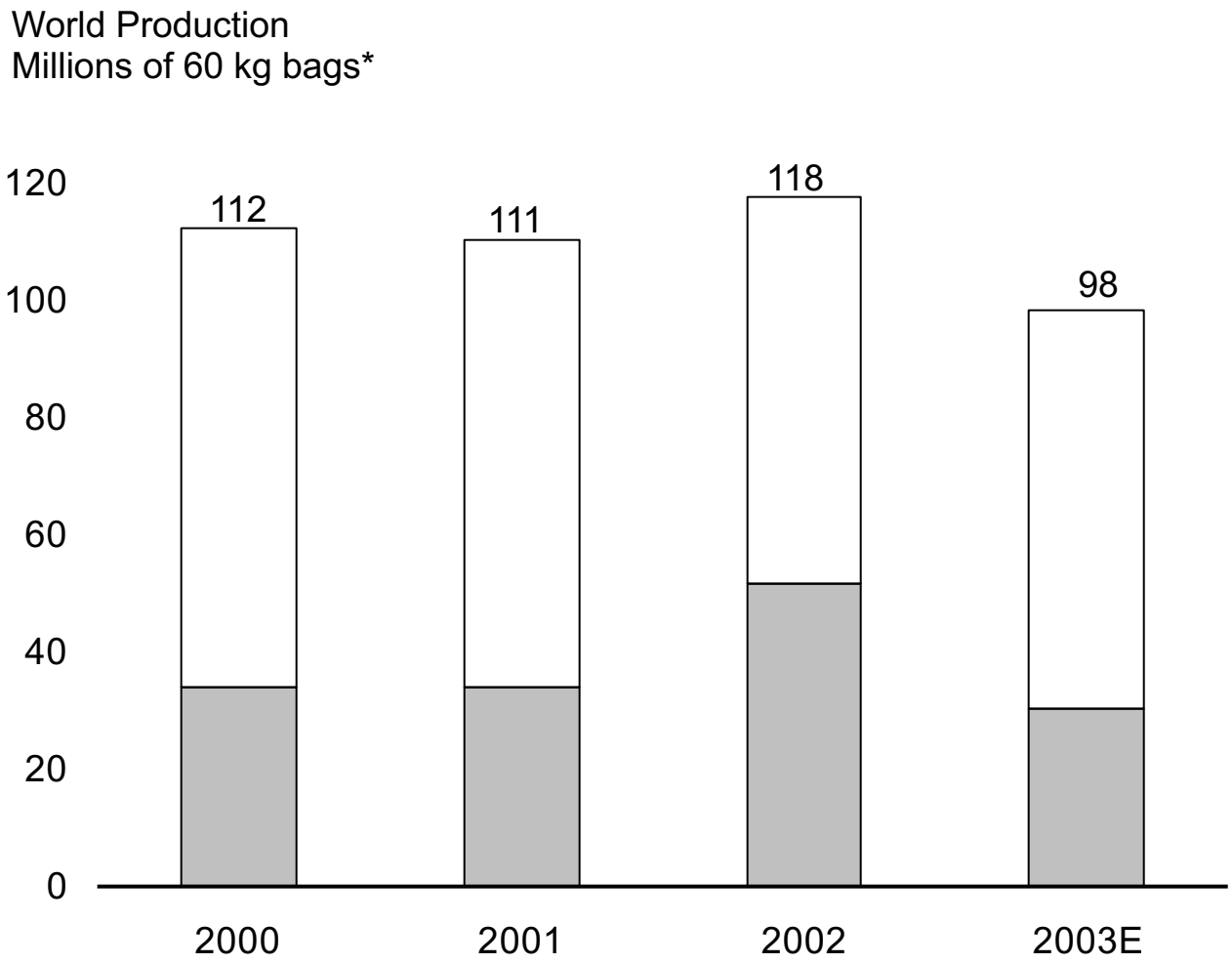
A SHORT-TERM PRICE REBOUND IS LIKELY GIVEN ESTIMATED BRAZILIAN PRODUCTION



- World demand for 2003 crop year is expected to be approximately 110 million bags

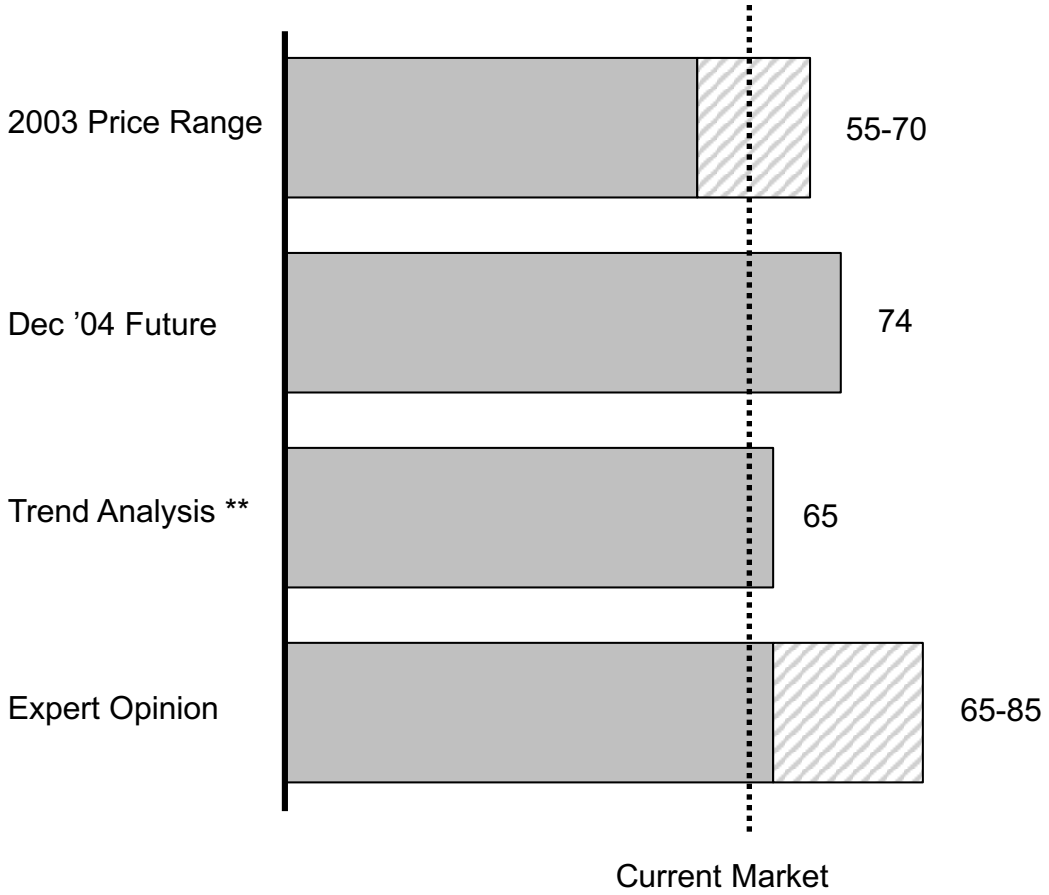
Brazil

* Crop years; historical data from USDA; 2003 estimate based on recent USDA and ICO estimates (best data available; seen as historically over-estimated)

Source: USDA; ICO

THE MEDIUM-TERM PRICE OF ARABICA IS EXPECTED TO AVERAGE 65 TO 85 CENTS PER POUND

Projected "C" market price
Dollar cents per pound

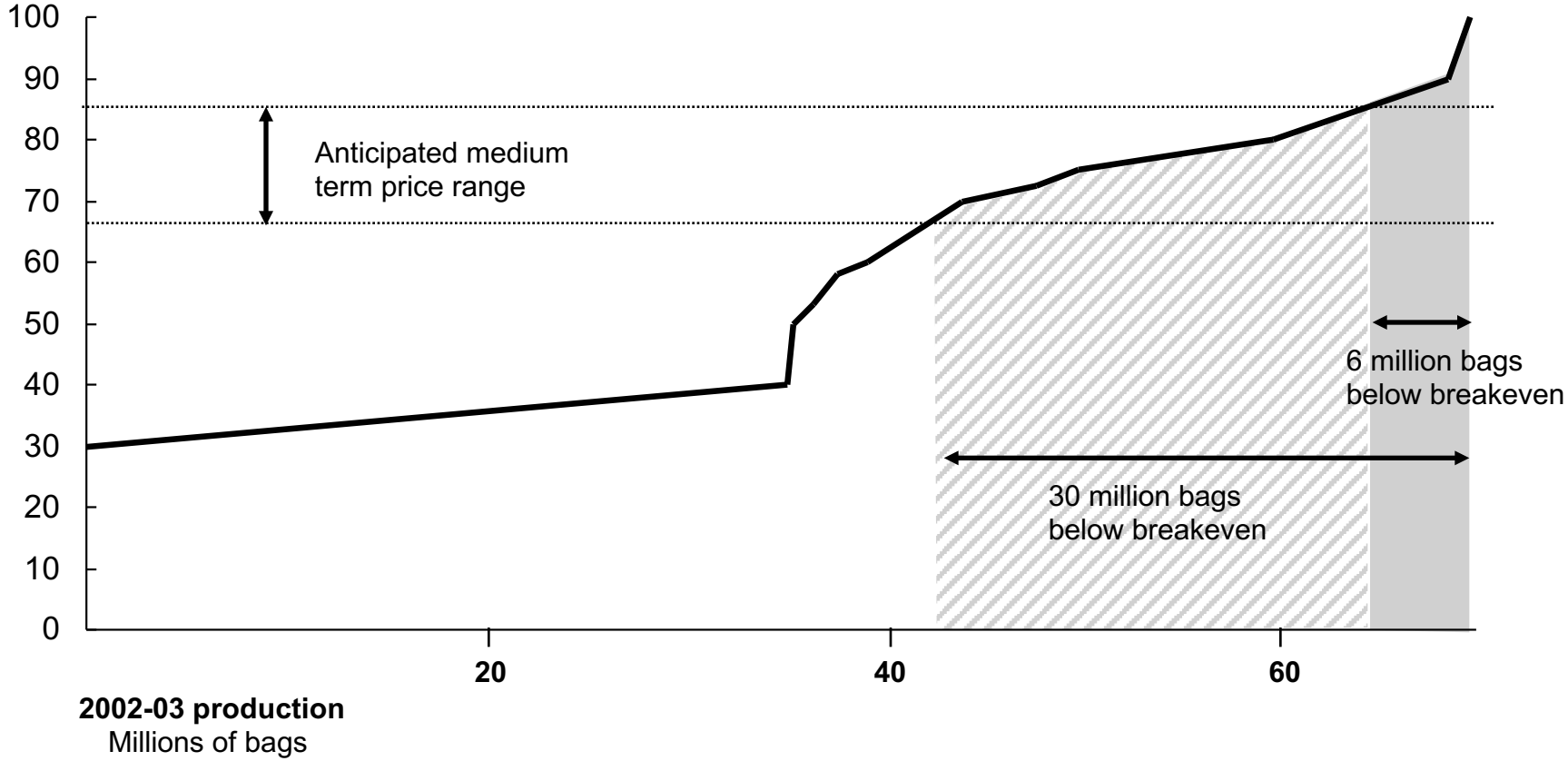


- Production deficit of ~8-13 million bags expected in 2003-04 crop year which will put upward pressure on price (ICO forecast)*
- World Bank predicts higher prices in 2005
  - 89.5 ¢/lb. for *Arabica* ("Other Milds")
  - 37.7 ¢/lb. for *Robusta*
- Judith Ganes Consulting predicts continued price recovery for the next 6-12+ months, followed by continued boom and bust cycles; cycles will be characterized by
  - Lows of 45-65 ¢/lb. (2-3 year periods)
  - Highs of 75-95 ¢/lb. (1-2 year periods)
  - Overall average of 70 ¢/lb.
  - Swings to $1.00-1.50/lb. may still occur but are expected to be brief
- December 2004 futures contract priced 17% higher than December 2003 futures contract

* Based on June 2003 ICO forecast
** ICO indicator price trend plus current "Other Milds" premium
Source: USDA; ICO; World Bank Global Economic Prospect, October 2001; Judith Ganes Consulting; team analysis

THIS PRICE RANGE WILL LEAVE MANY MARGINAL PRODUCERS UNPROFITABLE

Arabica FOB supply curve
Average cost (U.S. ¢/lb.), 1996-2001



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A RANGE OF POTENTIAL SOLUTIONS ARE ADVOCATED BY VARIOUS STAKEHOLDERS

Increase consumption

- **Increase Consumption** - Promote coffee consumption in producing countries and developing countries

Improve producer support

- **Risk Management** - Encourage producer use of risk management techniques (e.g., put options)
- **Market Information** - Facilitate transparency of information to producers (e.g., market information to create transparency of prices, improved crop forecasting to create transparency of supply)
- **Extension Services** - Improve coffee-related agricultural extension services (e.g., yield management)
- **Business Training** - Provide improved "business" awareness and ability
- **Vertical Integration** - Encourage producers to vertically integrate (e.g., into milling)
- **Roaster Relations** - Encourage long-term relationships between roasters and producers
- **Financing** - Provide access to financing (e.g., debt restructuring)

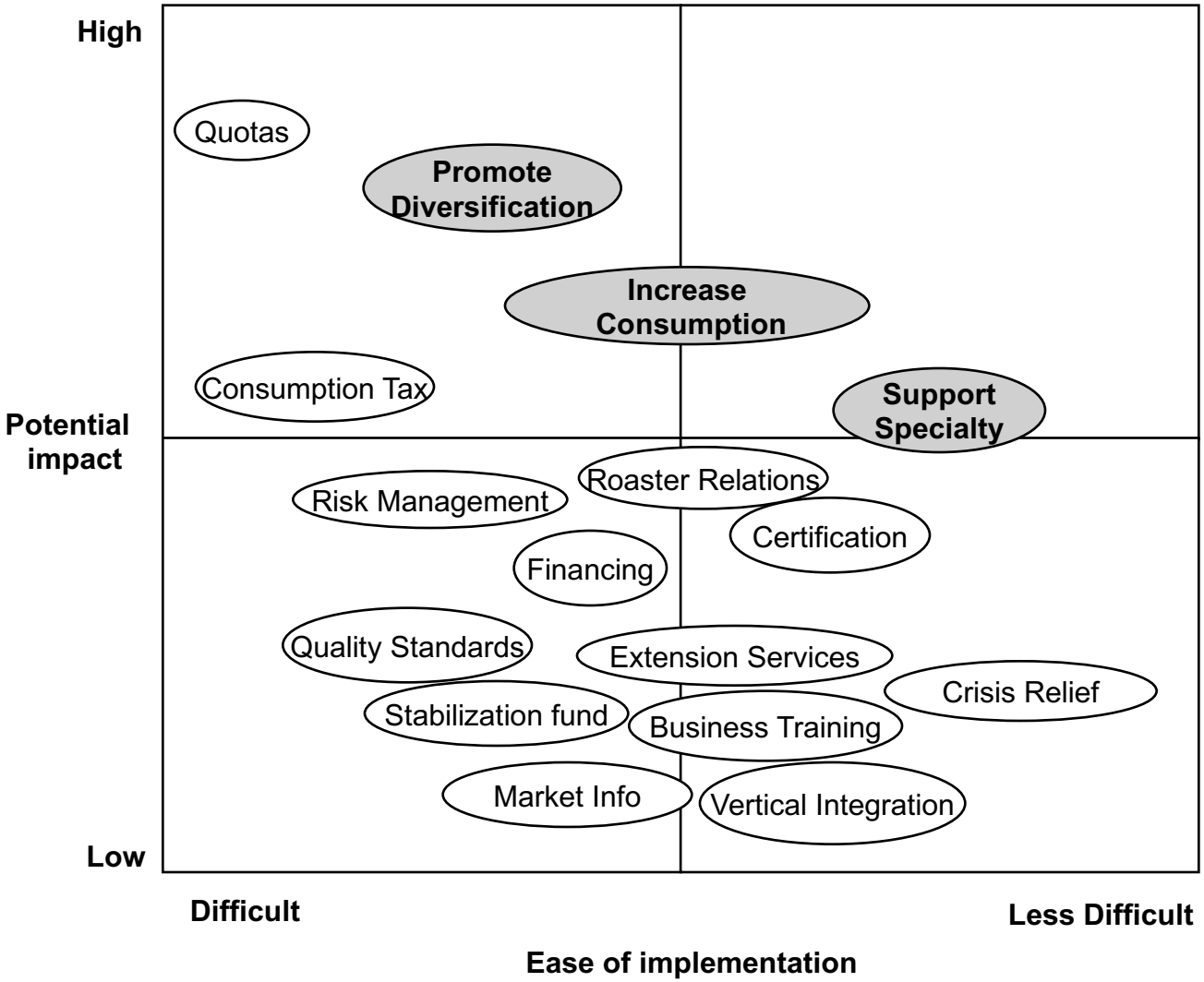
Transition in-crisis producers

- **Specialty** - Support move to improved quality ("specialty") coffee for advantaged producers
- **Certification** - Support certification and production of sustainable coffee for the specialty market (e.g., Fair Trade, organic, shade-grown/bird-friendly) and by large roasters (w/o price floor)
- **Diversification** - Promote partial or total diversification into alternative agricultural production and non-agricultural livelihoods (e.g., eco-tourism)

Apply macroeconomic policies

- **Quality Standards** - Institute quality resolutions to remove low-quality coffee from market
- **Stabilization Funds** - Provide long-term stabilization fund to minimize price volatility
- **Quotas** - Impose export quotas on coffee-producing countries
- **Tax Policy** - Reduce taxation on producers (country-specific)
- **Crisis Relief** – Provide relief for displaced coffee laborers
- **Protectionism** - Reduce US and European trade barriers to non-coffee products
- **Taxes** - Implement consumption tax on coffee

TECHNOSERVE BELIEVES THAT THREE MARKET SOLUTIONS OFFER THE HIGHEST POTENTIAL FOR SUSTAINABLE IMPACT



Potential Impact*
- Revenue - GDP increase
- Employment – Potential for job creation or preservation
- Scope – Broadly applicable or impacting a narrow segment
- Sustainability – Ability to preserve jobs/wealth over 5-20 year time frame

Ease of Implementation*
- Time Frame – Time required to generate economic growth
- Cost – Investment required
- Political Feasibility – Ability to reach stakeholder agreement and long term commitment
- Technical Feasibility – Complexity of solution and availability of local skills to implement

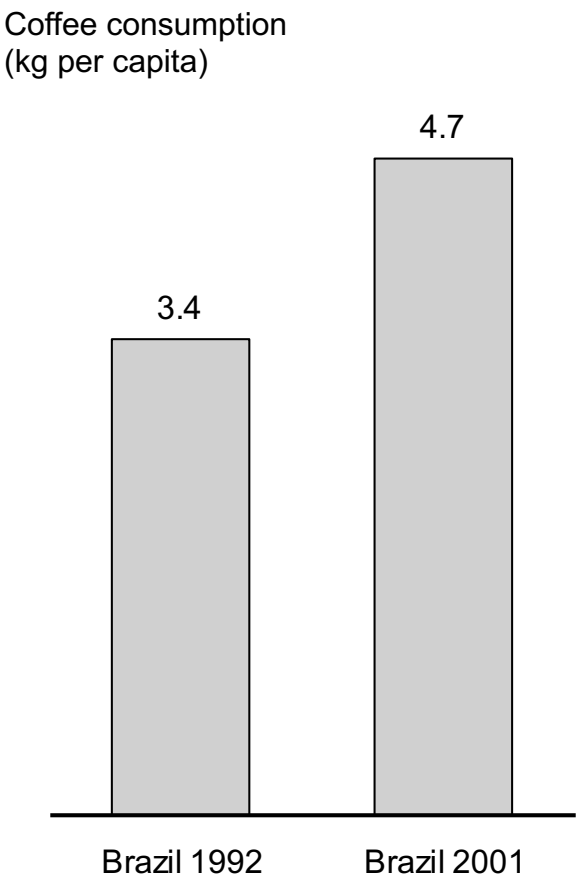
* These are qualitative assessments, opinions differ across the coffee industry on the potential relative impact of different solutions
Source: Team assessments

EACH OF THE THREE PRIORITY SOLUTIONS HAS THE POTENTIAL TO ASSIST DIFFERENT PRODUCER GROUPS

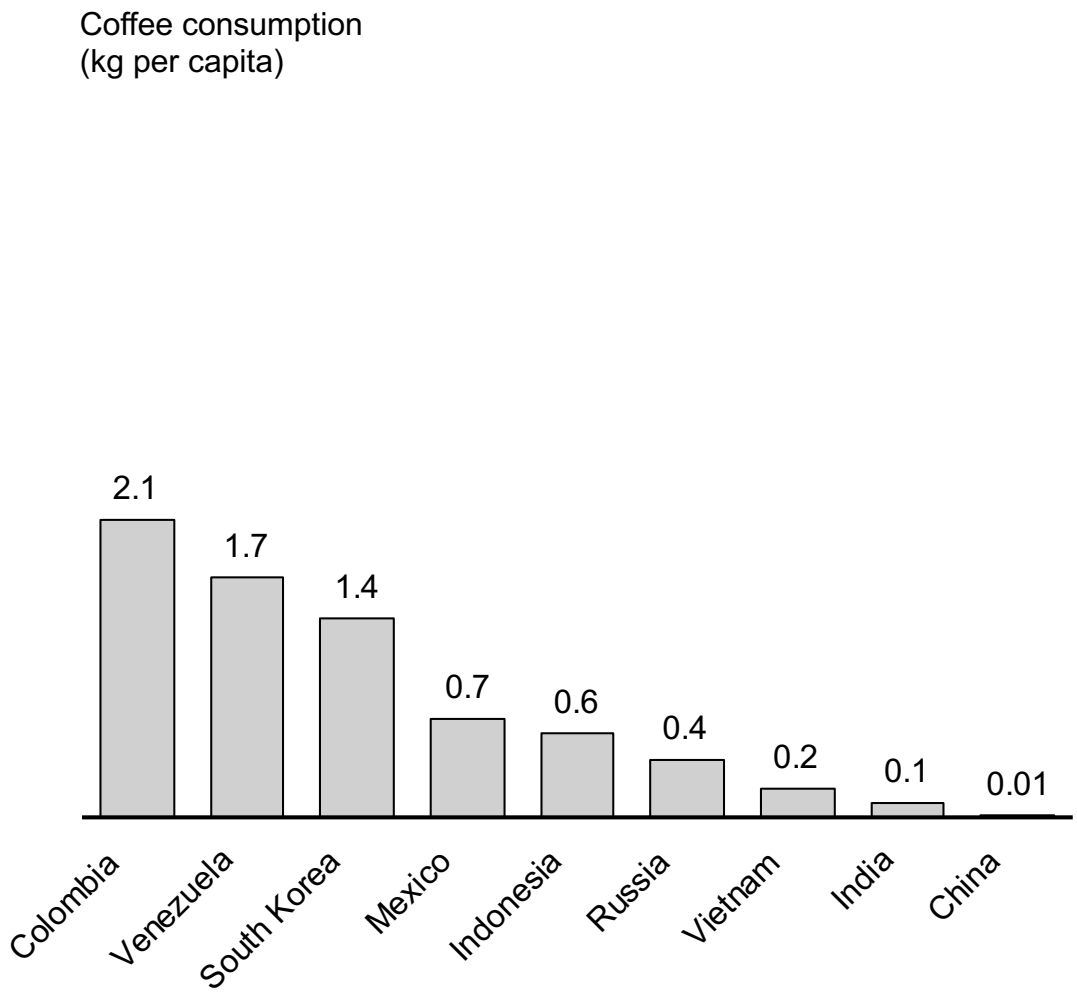
	Producer groups that benefit	Annual Revenue ($ millions)	One-time cost ($ millions)	Bags Impacted (millions)	Considerations
Increase consumption	• Producers with production costs below the long-term average price	340	60	5.0	• Brazil's success in increasing consumption may be difficult to replicate in countries without a history of high coffee consumption • Increasing demand may increase the volume sold without increasing the long-term price and unprofitable producers may not benefit
Support specialty	• Unprofitable producers that have the potential to differentiate and receive price premiums	100	50	2.6	• This solution is only applicable to a small portion of producers • Price premiums may erode with increasing supplies of specialty coffee
Promote diversification	• Unprofitable producers that do not have the potential to differentiate	500	250	5.0	• For some producers/regions it may be difficult to identify viable economic alternatives

INCREASING CONSUMPTION COULD BENEFIT COFFEE PRODUCERS

Coffee consumption in Brazil increased by 40% in a 10-year period



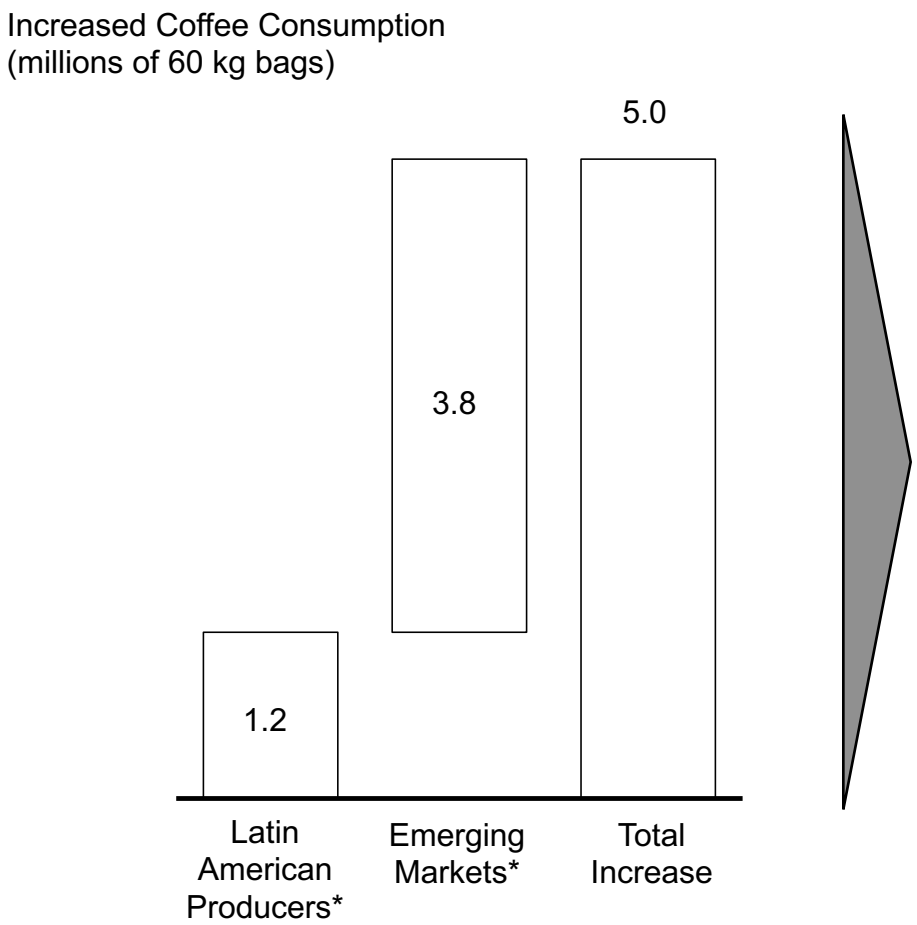
The potential may exist in other coffee producing countries and emerging market countries to replicate this increase in consumption



Source: Global Insight; ICO; team analysis

QUALITY AND MARKETING PROGRAMS COULD POTENTIALLY INCREASE CONSUMPTION

A 25% increase in per capita consumption in producer and emerging market countries would increase global demand by 5 million bags annually

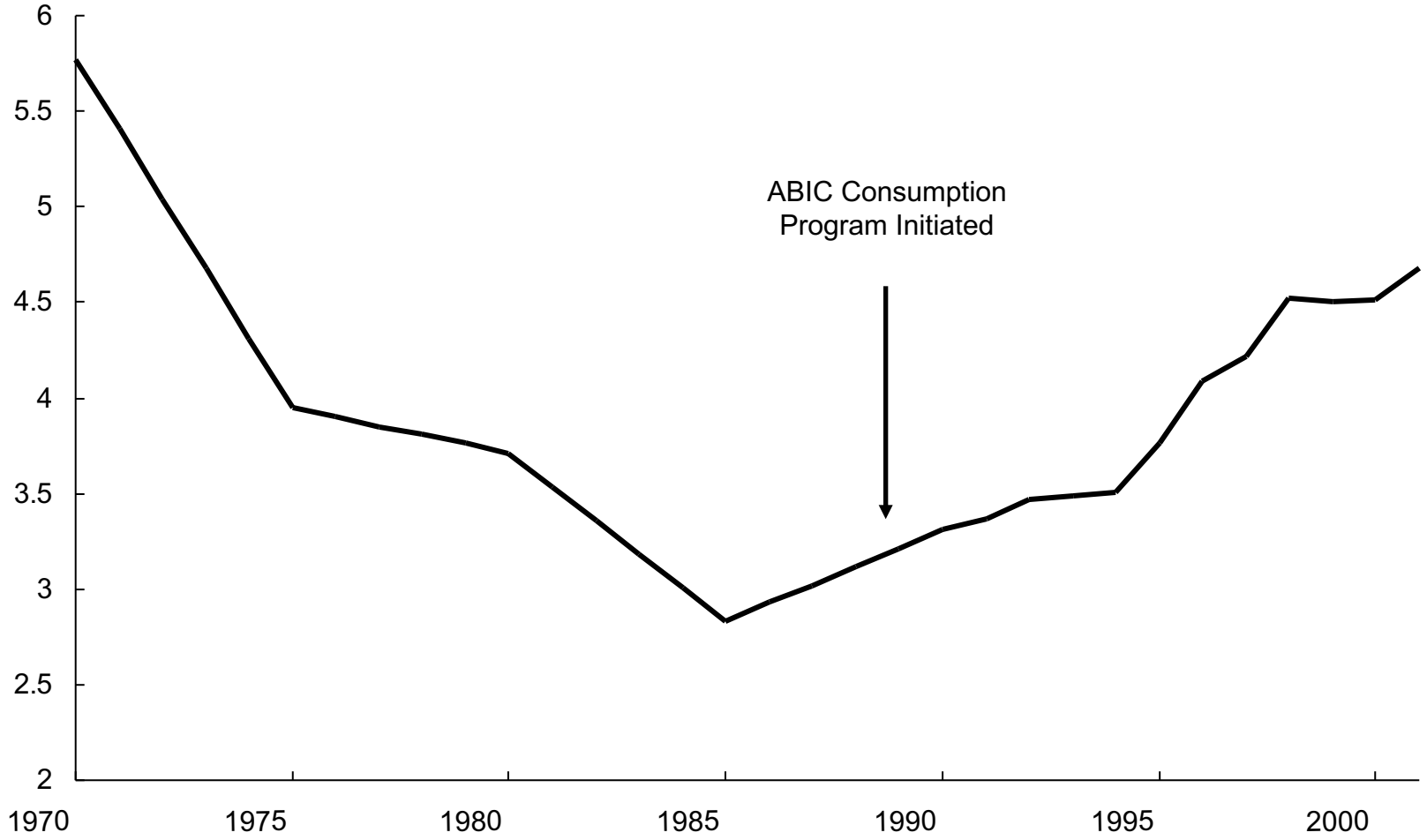


- Brazil's success in increasing consumption has been attributed largely to a quality program which included a purity certification
- A 25% increase in consumption in these countries may be feasible with programs similar to Brazil's
- Consumption programs would face varying cultural affinities for coffee, and a lack of scale in penetrating multiple smaller markets
- An increase in demand of this magnitude could increase consumption by 5 million bags which could increase revenue to coffee producers by $340 million annually **

* including 9 Latin America producing countries and 18 emerging market countries
**assuming a blended price of 52 cents/lb
Source:ICO, World Resources Institute, Team estimates

BRAZIL'S 40% CONSUMPTION INCREASE MAY BE DIFFICULT TO MEET IN COUNTRIES WITHOUT A HISTORY OF HIGH CONSUMPTION

Brazil's domestic coffee consumption
(kg per capita)



Source: ICO, ABIC

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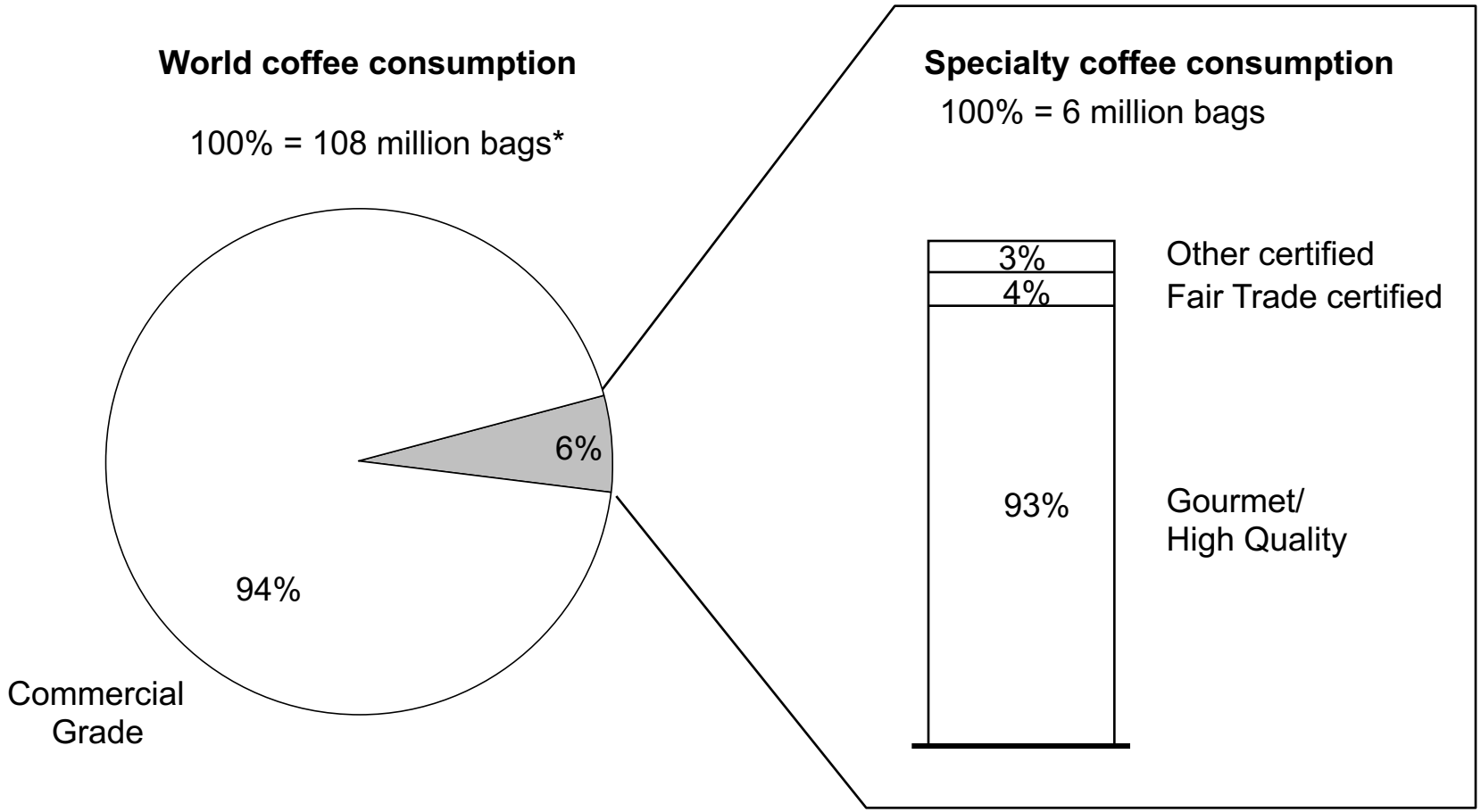
SPECIALTY COFFEE CURRENTLY REPRESENTS APPROXIMATELY 6 MILLION BAGS PER YEAR

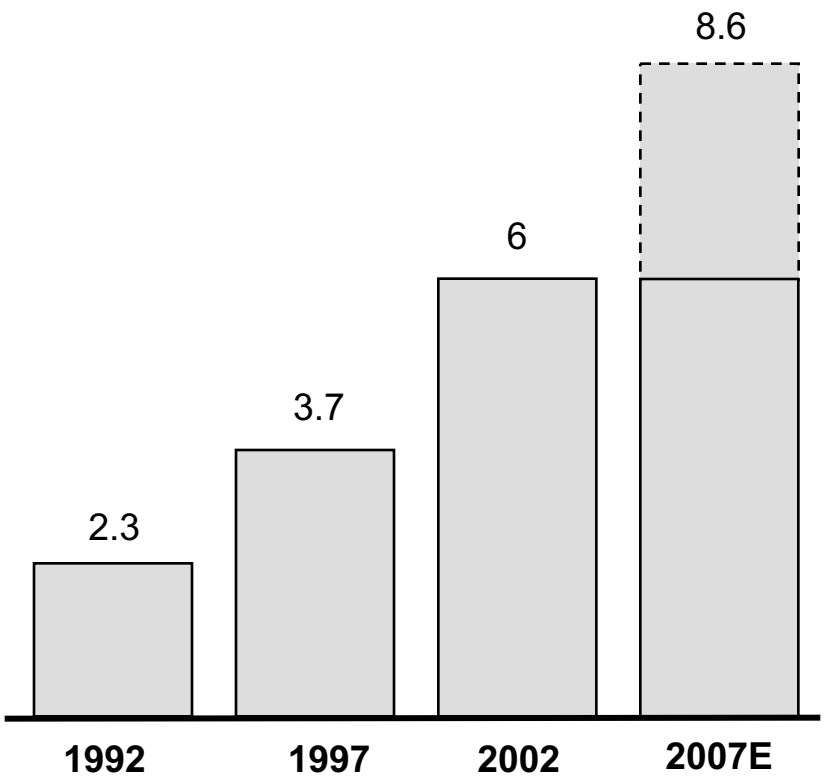


*1 bag = 60 kgs
Source:ICO; Daniele Giovannucci; Team estimate

ROASTERS WILL REQUIRE A CONSISTENT SUPPLY OF QUALITY SPECIALTY GRADE COFFEE TO MEET THE GROWING DEMAND

Demand may grow by 2.6 million bags by 2007*

Specialty consumption
(million of 60 kg bags)



However this demand growth will require consistent supplies of quality specialty coffee

- Roasters are concerned that at current prices producers will not invest to assure consistent supplies of specialty quality coffee
  - Starbucks is establishing regional buying offices offices
  - Nestle has established a specialty mill in Costa Rica
- 2.6 million bags would bring $100 million in additional revenue annually to coffee poducers assuming an average premium of 30 cents/lb
- This increase in specialty volume could potentially return 50,000 to 60,000 producers to profitability

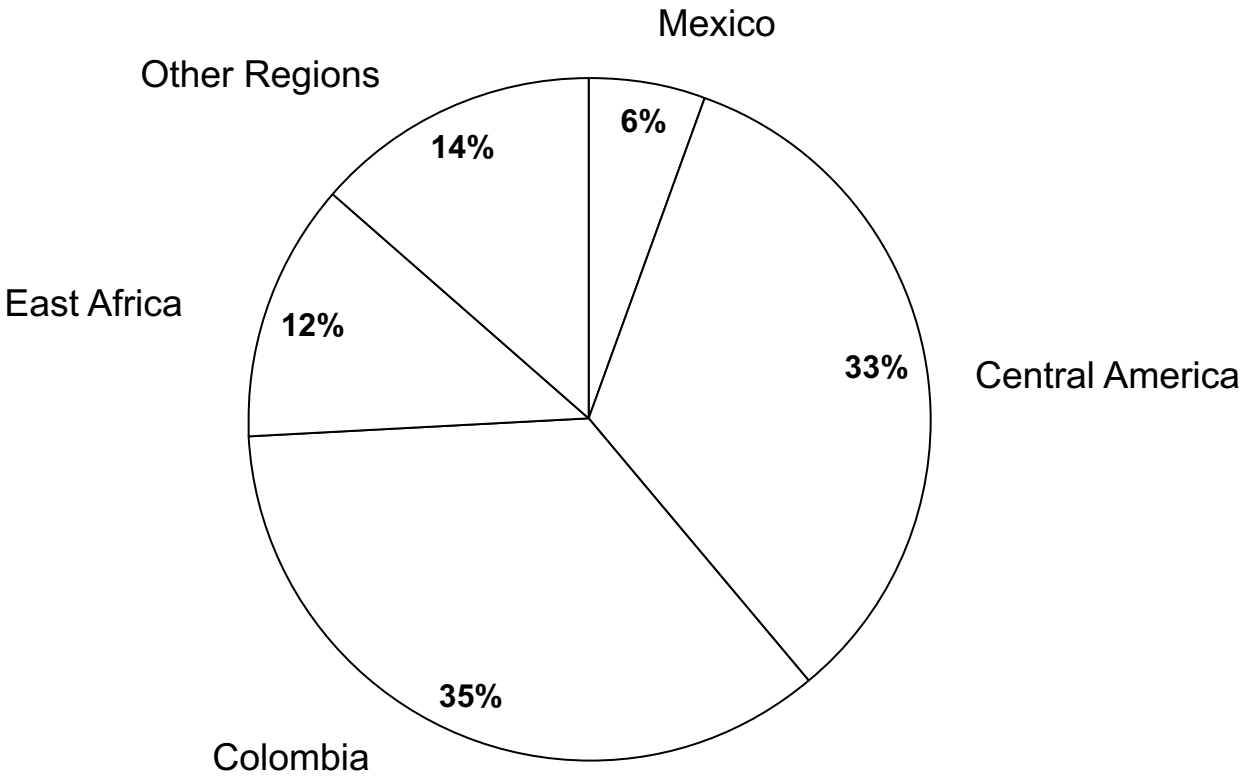
* Growth estimates for the specialty market range from 5 to 10% annually
Source:SCAA; Team estimate

EFFORTS SHOULD FOCUS ON HELPING COFFEE PRODUCERS MEET SPECIALTY COFFEE ROASTERS' DISCERNING STANDARDS

	Assess roaster needs	Identify potential farmers	Provide technical and business assistance	Facilitate access to financing	Provide industry-level support
Coffee Industry	• Define roaster needs based on market opportunities	• Participate in cupping competitions and auctions • Pursue longer-term relationships with farmers	• Give farmers feedback on quality • Provide hands-on support and training	• Provide financing to farmers with good track records	• Teach marketing seminars and participate in cupper trainings
NGOs	• Capture roaster needs • Communicate roaster needs to farmers	• Help organize cupping competitions and auctions • Identify high-potential farmers • Facilitate market linkages	• Improve yields and field processing • Improve wet milling facilities and techniques • Increase use of cupping	• Help access existing sources of financing • Set up fund for minor investments	• Coordinate marketing seminars and cupper trainings • Strengthen industry associations
Government		• Host coffee trade fairs		• Provide access to financing for working capital and equipment	• Address policy issues that constrain expansion of specialty coffee

SPECIALTY EFFORTS SHOULD FOCUS ON MEXICO, CENTRAL AMERICA, COLOMBIA, AND EAST AFRICA

2002 Specialty Coffee Production by Region



Source: SCAA

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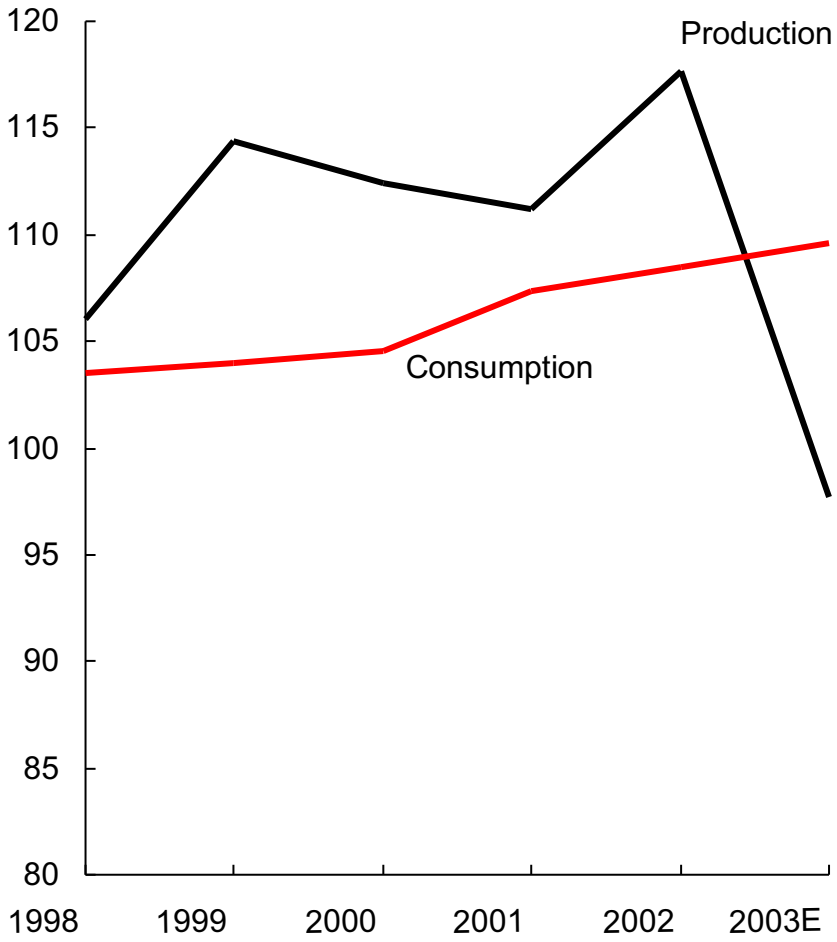
THE ECONOMIC VALUE OF AT LEAST FIVE MILLION BAGS OF COFFEE FROM HIGH COST PRODUCERS NEEDS TO BE REPLACED

Green coffee; millions of 60 kg bags



- Supply has exceeded demand in each of the last 5 years with an average surplus of 7 million bags
- The reduced harvest expected in Brazil this year, in combination with reduced global production due to low prices, is expected to result in an 8-13 million bag shortfall
- The 5 year average surplus including this year's anticipated shortfall is still 4 million bags
- If prices increase as a result of the shortfall, low cost producers and/or new entrants are likely to increase production
- TechnoServe estimates that at least five million bags will need to exit the market

* Real price index using 1992-2002 ICO indicator prices; all prices in 2002 dollars adjusted using World Bank Global Manufactured Unit Value index

Source: ICO; World Bank; *150 Years of Coffee*, Marcellino Martins and E. Johnston

HIGH-COST PRODUCERS WHO CANNOT DIFFERENTIATE NEED TO FIND ALTERNATE SOURCES OF INCOME

Competitive position

- Many high-cost producers of medium and low-grade coffee are unlikely to return to profitability
- Even if prices increase, low-cost producers and/or new entrants are likely to increase production

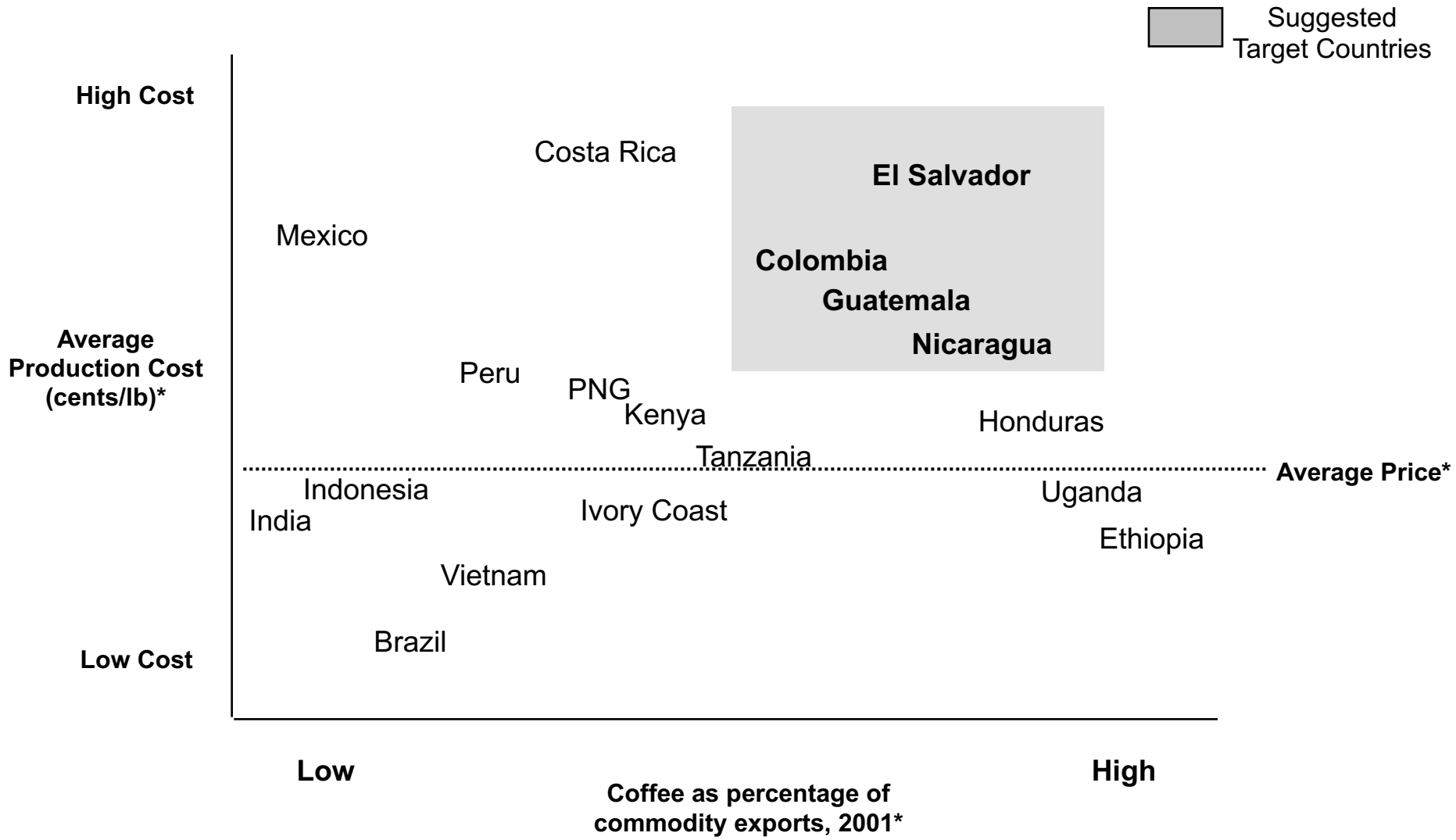
Potential resistance to diversification

- There is a long-standing cultural attachment to coffee that is difficult to break
- Experience with previous cycles has set the expectation that prices will rise in the future
- Producers who are operating close to profitability will be highly unlikely to consider diversification

Diversification Goal

- To diversify regional economies and reduce the dependence on coffee, TechnoServe estimates that alternative industries must be developed with the potential to generate $500 million in annual revenue

DIVERSIFICATION SHOULD BE FOCUSED ON COUNTRIES THAT ARE NOT COST COMPETITIVE AND ARE HIGHLY DEPENDENT ON COFFEE



* Cost positions are adjusted to reflect the relative prices differentials received by specific countries

Source: USDA; ICO Coffee Statistics, Sept 2002; Compete Coffee Coverage; TechnoServe; Economist Intelligence Unit; team analysis

DIVERSIFICATION SHOULD FOCUS ON INTRODUCING COMPETITIVE INDUSTRIES FOR REGIONS, NOT ON CROPS FOR INDIVIDUAL FARMERS

Approach	Rationale
• Promote economic diversification at a regional level rather than at a farm level	• A significant portion of the economic development in new industries is likely to take place outside existing coffee farms
• Work with coffee producers and local entrepreneurs	• Coffee producers may not be the pioneer entrepreneurs of new industries

SPECIFIC DIVERSIFICATION INDUSTRIES WILL BE IDENTIFIED FOR EACH REGION

Identify all potential industries

Prioritize based on a series of screens

To focus on set of opportunities

Industries suitable for a specific region

Market Potential

Country Competitiveness

Best Alternatives

1. Possible Industries
- Agro-climatic feasibility
- Import substitutes
- Value added processing of existing products
- New opportunities based on local advantage

2. Attractive Opportunities
- Potentially attractive domestic or export market
- Size of investment required and potential margins available

3. Potential for Success
- Definable competitive advantage
- Technical complexity
- Infrastructure investments required
- Feasibility of establishing market links

NO SINGLE INDUSTRY IS LIKELY TO BE IDEAL FOR A TARGETED REGION

Example Industries	Advantages	Drawbacks
Fruits and vegetables	• Large and growing market • Generator of employment and quick cash flow	• Need for good roads to markets • Investment in infrastructure required • Perishable product
Ornamental plants	• Growing market • Generator of employment and quick cash flow	• Highly demanding market • Large investment in infrastructure required • Delicate product
Cocoa	• Large market and good prices • Production, processing, and marketing similar to that for coffee	• Commodity crop subject to sharp fluctuations in prices • Risk of oversupply in several years
Cattle	• Viable market for meat and dairy products	• Need for relatively gentle slopes • Environmental damage due to clearing of land
Forestry and timber	• Good market opportunities, especially for tropical hardwoods • Positive impact on environment	• Limited generator of employment • No short-term cash flow
Eco-tourism	• Growing market • High-value industry	• Relatively small industry • Longer-term investment

SUCCESSFUL IMPLEMENTATION WILL REQUIRE THE COORDINATED INVOLVEMENT OF THE PRIVATE SECTOR, NGOs AND GOVERNMENTS

	Select promising industries	Identify business opportunities	Provide technical and business assistance	Facilitate access to financing	Provide industry-level support
Private Sector	• Provide support in selection of industries	• Explore new business opportunities	• Share best practice examples	• Provide financing to entrepreneurs	• Teach marketing and technical seminars • Help build industry associations
NGOs	• Assess market potential and country competitiveness for each industry • Help select 3 to 5 industries per country	• Identify entrepreneurs • Facilitate market linkages	• Develop business plans • Strengthen operations and quality control • Provide support in management and accounting	• Help access existing sources of financing • Set up fund for minor investments	• Coordinate marketing and technical seminars • Strengthen industry associations
Governments	• Participate in selection of promising industries	• Host industry trade fairs		• Provide access to financing for investments in equipment	• Address policy and infrastructure issues that constrain industry growth

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- Overview of the Current Crisis
- Potential Solutions
- Likely Developments
- **Next Steps**